Chenshuhao (Cody) Qin

Los Angeles, CA | (213) 922-0925 | cody2363264403@gmail.com | github.com/CodyQin | linkedin.com/in/chenshuhao-qin

Education

University of Southern California (USC) Los Angeles, CA
Master of Science in Computer Science *Jan. 2026 – Dec. 2027 (Expected)*

- **Coursework**: Machine Learning, Deep Learning, Natural Language Processing, Data Structures & Algorithms, Operating Systems, Computer Architecture, Computer Networks, Predictive Modeling

Duke University; Duke Kunshan University Durham, NC / Kunshan, China
Bachelor of Science in Computer Science; B.S. in Computation and Design *Aug. 2021 – May. 2025*

- **Honors**: **Apple WWDC23 Swift Student Challenge Winner** (1 of 13 in China), Provincial Innovation Grant Recipient ($3k), DKU Student Ambassador

Skills

Programming: Python, TypeScript/JavaScript, Java, SQL (MySQL, PostgreSQL), Swift, C
AI & Machine Learning: PyTorch, TensorFlow, LoRA Fine-tuning, Prompt Engineering, Hugging Face, Pandas, NumPy
LLM & Agents: LLM Agents (AgentScope, LangChain, LlamaIndex), Retrieval-Augmented Generation (RAG), Model Serving (vLLM), Embeddings & Vector Search (Milvus), Ray
Infrastructure & Tools: Kubernetes, Docker, CI/CD, Git, Linux, REST APIs (FastAPI, Flask), React, Redis, Object Storage (OSS/S3), AWS, PyTest

Work Experience

Alibaba Cloud (Top-4 Global Cloud Provider) Hangzhou, China
AI Engineer Intern *Jun. 2026 – Aug. 2026*

- Collaborated with Alibaba's **Tongyi Lab** to productize **Data-Juicer Agents** into natural-language-driven multimodal data pipelines (cleaning, filtering, deduplication) for the broader **Qwen** ecosystem, replacing script-heavy workflows on the Apsara Data Lake platform.
- Built an agent-to-**Ray** distributed execution pipeline over private-cloud object storage (OSS) and Kubernetes — **2,793 records, 16 operators, ~5 minutes** — and landed the core capability upstream through a merged PR.
- Designed a three-layer context-control architecture (tool-output compaction, memory compression, hard token budgets) cutting single-tool context payloads by **82%** and enabling stable multi-turn agents on 30k-token models.
- Shipped a full-stack agent workspace (**React / FastAPI / MySQL**, REST APIs) with persistent sessions, execution tracing, conversation rollback, and multi-instance isolation; delivered **4 production container-image releases**.
- Contributed **5 merged PRs** (PyTest unit coverage, CI green) to the 7k⋆ Data-Juicer open-source ecosystem across multimodal ingestion, distributed Ray execution, and context optimization; 1 enhancement proposal adopted by maintainers.

Blueberry AI Burlingame, CA (Remote)
AI Engineer Intern *Nov. 2025 – Jan. 2026*

Built a **Milvus + MySQL** feature-extraction and auto-tagging pipeline batch-indexing **200K+ images** for landmark / IP / SKU retrieval; benchmarked **DINOv3, Qwen-VL, and BGE-VL** under a unified recall / precision / latency framework for private deployment; delivered a YOLO + **Milvus** vector search + Gradio retrieval POC validated for commercial use.

Tsinghua Sichuan Energy Internet Research Institute Chengdu, China
Algorithm Engineer Intern *Sep. 2025 – Oct. 2025*

Engineered a **DeepLabV3+ (PyTorch)** semantic-segmentation model for microgrid fault diagnosis reaching **92% accuracy** across complex field scenarios; owned the full data lifecycle of **500+ annotated images** from preprocessing to deployment, anchoring winning bids for major energy-infrastructure projects.

Leiting Games (G-bits) Shenzhen, China
AI R&D Intern (AIGC & Automation) *May. 2025 – Aug. 2025*

Fine-tuned **SDXL / Flux LoRA** models on **Kohya**, shipping 30+ ad assets (80% adoption) that drove **117K+ installs** and **¥24.6M billings (196% ROI)**; led a 5-person team scaling creative output **3**× with 15–20% higher material ROI; built a **98.85%-accurate** RPA monitor catching 1,000+ private-server violations (2025 Best Practice).

Projects

ChatDKU — RAG Campus Q&A Assistant | *Python, LlamaIndex, LangChain, Redis* Jul. 2024 – May. 2025

Co-founded a university-adopted **Retrieval-Augmented Generation (RAG)** Q&A agent indexing **12,000+ webpages** over a privately deployed LLM; implemented **Hybrid Search (BM25 + vectors)**, **HyDE**, multi-level **Redis** caching, and **SSO** web deployment, attaining **ROUGE 81.1% / BLEU 60.2%**; authored the design as a DKU Signature Work.

Publications

Privacy-Preserving Room Occupancy Estimation Using Federated Analytics of BLE Packets — **ACM SenSys 2024** (Demo Track, second author), pp. 889–890; **95%** real-world accuracy via federated learning over BLE.